

Aditi Kathalay

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Work Experience

L&T Finance | *Manager, AI Engineer*
Mumbai/Pune, India

Jun 2024 – Present

- Currently developing **Helios CAM**, a one-stop credit underwriting solution integrating **banking analysis, bureau analysis, loan deduplication, and multi-GST consolidation** to streamline end-to-end credit assessment.
- Key Helios flagship modules:** SME Triangulation for cross-verifying GST, income, and banking data; HL FSA for automated ITR and financial statement analysis; and PL Bureau Analyser for comprehensive credit bureau assessment.
- Built a **Machine Learning-driven Gemini Lifestyle Index Rating model** to assess housing quality, reducing low-quality loan approvals and saving field officers' time; delivered the complete **full-stack AI solution across frontend, backend, and APIs**, showcased at **RAISE 2024**.

L&T Finance | *Data Science Intern*
Mumbai, India

Jan 2024 – May 2024

- Developed a **Treasury Forecasting model** using Prophet, improving product-level financial accuracy for treasury planning.

Skills

AI/ML: Machine Learning, Agentic AI, Tool/Function Calling, LangChain, LangGraph, LLMs, MCP, RAG

Programming: Python, R, SQL

Frameworks & Tools: Flask, FastAPI, Power BI, Git, Excel

Cloud: Google Cloud Platform, AWS

Certifications

Python (Google–Coursera), SQL (UC Davis–Coursera), Tableau (Scaler)

Projects

Personalized Skincare and Beauty Assistant | *Multi-Agent, MCP, RAG*

Ongoing

- Building a multi-agent conversational bot for skincare, haircare, and makeup, providing personalised recommendations using RAG with curated product data.
- Integrated **Model Context Protocol (MCP)** with e-commerce platforms including Amazon, Flipkart, and Nykaa for product discovery and checkout.

AI-Powered Travel Planner | *Multi-Agent, Streamlit, LangGraph*

2025

- Developed an interactive multi-agent travel planner (GitHub) generating itineraries, hotels, attractions, weather, and cost breakdowns.
- Designed LangGraph agent workflows to prevent recursion loops and support scalable API integration.

Diabetic Retinopathy Detection and Stage Classification | *Python, OpenCV, Perceptron, CNN*

Aug 2023

- Applied GRAHAM preprocessing and trained CNN + Perceptron models to classify Diabetic Retinopathy stages, visualizing key retinal features to improve model interpretability.

Education

M.Sc, Data Science and Big Data Analytics, Maharashtra Institute of Technology, Pune

Aug 2022 – Jun 2024

Gold Medalist, GPA: 9.6

B.Sc, Computer Science, Savitribai Phule Pune University, Pune

Aug 2019 – Jun 2022

Gold Medalist, GPA: 9.7

Achievements

- Project Helios** won the **NASSCOM AI Game Changers Award 2026** for **L&T Finance** in the **BFSI sector**, recognizing its impact in AI-driven credit underwriting.
- Served as **Master of Ceremonies** for the prestigious **L&T Finance Star Awards 2025** and received a company award for outstanding hosting.
- Presented research paper “*Towards Early Detection: A Deep Learning & AI Approach to Diabetic Retinopathy Diagnosis*” at the **International Conference on Machine Learning & Intelligent Systems (ICMIS 2023)**.